

ID5104: Stochastic Calculus for Finance

Final exam *checklist*

No proofs will be asked in the exam. It is sufficient if the statement and implications of the theorems mentioned below are clearly understood.

Specific formulae and notation which have lots of symbols and integrals and are hard to remember will be provided in the question paper.

Standard definitions and results will **NOT** be provided in the question paper – you are expected to know them.

It is recommended to go through Assignments 2-6 carefully (including the theoretical parts).

Not included for the final exam:

1. Random walk theory (first passage times, maximum process for random walk, reflection principle etc.) However, the binomial tree model for discrete stock price process is included.
2. American derivatives. But simple facts about stopping times are included in the exam (see Section I below).
3. Questions related to programming and numerical methods.

Section I: Math background

Probability basics

- Shreve I, Chapter 2
- Shreve I, Chapter 3, Sections 3.1, 3.2
- Shreve I, Chapter 4, Section 4.3- Shreve I, Chapter 4, Section 4.3
- Shreve II, Chapter 2

- ☐ Sigma algebras and Filtrations
- ☐ Conditional Expectations
- ☐ Change of measure: Radon-Nikodym derivative
- ☐ Radon-Nikodym derivative that shifts the mean of a standard normal random variable
- ☐ Markov processes
- ☐ Martingales
- ☐ Stopping times - Definition, optional sampling theorem

Brownian motion or Wiener process

- Shreve II, Chapter 3, Section 3.3, 3.4, 3.5, 3.7

- ☐ Brownian motion: definition, distribution, filtration
- ☐ Markov property and Martingale property
- ☐ Quadratic variation of Brownian motion
- ☐ Reflection principle for Brownian motion
- ☐ Joint distribution of Brownian motion and its max-to-date process

Stochastic Calculus

- Shreve II, Chapter 4, Sections 4.1, 4.2, 4.3, 4.4, 4.6

- ☐ Ito integrals: definition
- ☐ Properties of Ito integral in Theorem 4.3.1
- ☐ Ito processes
- ☐ Ito-Doeblin formula for Brownian motion and Ito processes
- ☐ Ito integral for deterministic integrand (Theorem 4.4.9)
- ☐ Multivariable Stochastic calculus
- ☐ Levy's theorem (Theorem 4.6.4)

Stochastic and Partial differential equations

- Shreve II, Chapter 6, Sections 6.1, 6.2, 6.3, 6.4

- ☐ Stochastic differential equations
- ☐ Martingale property of conditional expectations of functions of Markov processes
- ☐ Feynman-Kac and discounted Feynman-Kac theorems
- ☐ The 'Feynman-Kac technique': computing the Ito-Doeblin differential of a discounted risk-neutral price process and setting the dt term to zero since the discounted risk-neutral price process is a martingale with respect to the risk-neutral measure

Section II: Applications to Derivatives Pricing

Binomial tree model for asset pricing - Shreve I, Chapter 1

- ☐ The binomial tree model for stock prices
- ☐ European call and put options in discrete time
- ☐ The wealth process of the Delta-hedging portfolio for a short position in a European option
- ☐ Risk-neutral pricing of European options using Delta-hedging

Black-Scholes-Merton model for asset pricing - Shreve II, Chapter 4, Section 4.5

- ☐ The geometric Brownian motion model for stock prices in continuous time
- ☐ The hedging portfolio for a short position on a European call option in continuous time
- ☐ Deriving the Black-Scholes-Merton differential equation for the price of a European call option by equating the evolution of Delta-hedging portfolio process and option price process
- ☐ Properties of the Black-Scholes-Merton solution for price of European call option
- ☐ The Greeks - Delta, Theta, Gamma
- ☐ Forward contract prices and Put-Call parity

Application of general ideas of stochastic calculus to derivative pricing

- Shreve II, Chapter 5, Sections 5.1, 5.2, 5.3, 5.4

- ☐ Girsanov's theorem for one dimensional Brownian motion
- ☐ Empirical probability measure and risk-neutral probability measure
- ☐ **Risk-neutral pricing formula for assets**
- ☐ Deriving the Black-Scholes-Merton solution for a European call using the risk-neutral pricing formula and properties of the Wiener process
- ☐ Martingale representation theorem and existence of Delta-hedging portfolio
- ☐ Multi-dimensional market model based on multidimensional Brownian motion
- ☐ Multidimensional Girsanov's theorem and Martingale representation theorem
- ☐ Conditions for existence of a risk-neutral measure: market price of risk equations
- ☐ Conditions for uniqueness of risk-neutral measure: the Delta-hedging equations
- ☐ Absence of arbitrage and the first fundamental theorem of asset pricing
- ☐ Completeness of a multidimensional market model and the second fundamental theorem of asset pricing

Interest rate models and interest rate dependent derivatives

- Shreve II, Chapter 6, Section 6.5

- Shreve I, Chapter 6, Sections 6.1, 6.2, 6.3

- ☐ Elementary continuous time short rate models: Vasicek, Hull-White, Cox-Ingersoll-Ross
- ☐ Binomial model for the interest rate process and the discount process
- ☐ Zero coupon bonds and their risk-neutral pricing
- ☐ Coupon-paying bonds and their risk-neutral pricing

Interest rate models and interest rate dependent derivatives

- Shreve II, Chapter 6, Section 6.5

- Shreve I, Chapter 6, Sections 6.1, 6.2, 6.3

- ☐ Forward contracts on interest rate dependent assets
- ☐ Risk-neutral price and m-forward price of an interest rate dependent asset
- ☐ Original interest rate and m-forward interest rate
- ☐ Interest rate contracts (Theorem 6.3.5 in Shreve I)
- ☐ m-period Interest Rate Swap, m-period swap rate (Definition 6.3.6 in Shreve I), risk-neutral pricing
- ☐ m-period Interest Rate Cap and m-period Interest Rate Floor
- ☐ Relation between risk-neutral prices of swaps, caps and floors
- ☐ Swaplets, Caplets and Floorlets